

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fifth Semester of B. Pharm. Examination
University Theory Examination December 2017
PH318 BIOPHARMACEUTICS AND PHARMACOKINETICS

Date: 04.12.17, Monday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Majority of drugs bind to extravascular tissues, the order of binding is:
[A] Liver > Kidney > Lung > Muscles
[B] Kidney > Lung > Muscles > Liver
[C] Lung > Muscles > Liver > Kidney
[D] Muscles > Liver > Kidney > Lung
 2. Which of the following refers to “Metabolism”?
[A] The process of substance entering the body.
[B] The dispersion of the substances throughout the body.
[C] The ongoing interrelated series of chemical interactions taking place in living organisms that provide the energy and nutrients needed to sustain life.
[D] The process of discharging waste matter from the tissues or organs.
 3. Which of the following route of administration lack “Permeation of drug” step?
[A] Oral
[B] Subcutaneous
[C] Intramuscular
[D] Intravenous
 4. Which of the following reason is the main cause responsible for determining bioavailability?
[A] Rheological parameters of blood
[B] Quantity of food intake
[C] Extent of absorption and hepatic first-pass effect
[D] Glomerular filtration rate

5. Metabolic transformation (Phase-I) is:
 - [A] Acetylation and methylation of substances
 - [B] Transformation of substances due to oxidation, reduction or hydrolysis
 - [C] Glucuronide formation
 - [D] Binding to plasma
6. Half-life ($t_{1/2}$) is the time required to:
 - [A] Change the amount of a drug in plasma by half during elimination
 - [B] Metabolize a half of an introduced drug into the active metabolite
 - [C] Absorb a half of an introduced drug
 - [D] Bind the half of an introduced drug to plasma proteins
7. Compartment models are also known as _____.
 - [A] Physiological models
 - [B] Empirical Models
 - [C] Distributed models
 - [D] Realistic models
8. The penetration of drug in infants is _____ than adults due to poorly developed Blood-Brain Barrier.
 - [A] Lower
 - [B] Higher and then lower
 - [C] Higher
 - [D] Not getting changed
9. Which of the following characteristics is most likely to be associated with a high apparent volume of distribution?
 - [A] Extensive binding to tissue constituents
 - [B] Distribution into total body water
 - [C] Extensive binding to plasma protein
 - [D] Penetration across the blood: brain and blood: testis barriers
10. _____ is a best test to quantitatively assure the biological availability of drug from the formulation.
 - [A] Disintegration test
 - [B] In-vitro dissolution test
 - [C] Friability test
 - [D] Hardness Test
11. Following is not the approach to promote the crossing of drug through Blood Brain Barrier:
 - [A] Use of Drug carriers
 - [B] Use of Mannitol
 - [C] Intensification of perfusion rate
 - [D] Use of DMSO (Dimethyl sulfoxide)
12. In multiple dose study to ensure steady-state, the drug should be administered for _____ elimination half-lives before collecting the blood samples.
 - [A] 4-6
 - [B] 5-6
 - [C] 6-7
 - [D] 7-8

13. Drugs administered _____ may diffuse directly into CNS.
 [A] Intra-nasally
 [B] Intra-dermally
 [C] Intra-muscularly
 [D] Orally
14. When systemic availability of orally administered drug is compared with an oral standard of the same drug, it is referred to as _____.
 [A] Absolute Bioavailability
 [B] Partial Bioavailability
 [C] Relative Bioavailability
 [D] Actual Bioavailability
15. Urinary excretion studies works on the principle that the urinary excretion of _____ drug is directly proportional to plasma concentration of drug.
 [A] Altered
 [B] Metabolized
 [C] Ionized
 [D] Unchanged
16. The ratio of maximum safe concentration to minimum effective concentration of the drug is called as _____.
 [A] MEC
 [B] Therapeutic window
 [C] MSC
 [D] Dosage Regimen
17. A decrease in the drug metabolizing ability of an enzyme is called as _____.
 [A] Enzyme mobilization
 [B] Enzyme Induction
 [C] Enzyme Restriction
 [D] Enzyme Inhibition
18. Wagner Nelson method applies only to drugs that follow:
 [A] Multicompartment kinetics
 [B] One compartment kinetics after oral administration
 [C] One compartment kinetics after iv infusion
 [D] One compartment kinetics after iv bolus
19. _____ barrier is located not at the capillary endothelium level but at sertoli-sertoli cell junction.
 [A] Blood-Testis
 [B] Placental
 [C] Blood-cerebrospinal fluid
 [D] Blood-brain
20. Non- linear kinetics is also known as _____.
 [A] First order kinetics
 [B] Mixed order kinetics
 [C] Zero order kinetics
 [D] Second order kinetics

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Define the following: 05
- (i) Onset of action
 - (ii) In-vivo In-vitro Correlation (IVIVC)
 - (iii) Therapeutic equivalence
 - (iv) Drug Disposition
 - (v) Drug Absorption
- B** Describe Biopharmaceutical Classification System (BCS) in brief. 05
- C** State the various Patient related factors affecting drug absorption from its dosage form and explain any one in detail. 05
- Q 3** Attempt any **TWO** of the following
- A** Simplify the effect of hepatic blood flow rate-limited clearance of drug in detail. 05
- B** Recognize the causes of non-linearity in drug absorption and drug excretion and explain it in brief. 05
- C** Differentiate: Passive transport and Active transport. 05

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Discuss in detail “volume of distribution” with its significance. 05
- B** Enlist the factors affecting distribution of the drugs. Explain effect of organ/tissue size and perfusion rate in distribution of drug in detail. 05
- C** Write a short note on Blood–Brain barrier. 05
- D** (i) What is flip-flop phenomenon? When is it observed? 03
(ii) Define: AUC and T_{max} . 02
- E** Enumerate the methods for enhancement of bioavailability and explain any one method in detail. 05
- F** Comment on following:
- (i) Phase-I reactions are called as functionalization reaction. 03
 - (ii) Glucuronidation is the commonest and most important of all phase II reactions. 02
- Q 5** Attempt any **FOUR** of the following
- A** Explain Wagner – Nelson method for estimation of absorption rate constant (K_a) in detail. 05
- B** (i) Define following terminology:
(a) Bioavailability (b) Drug Excretion 02
(ii) Enlist Phase I reactions for biotransformation of drugs. 03
- C** Explain the various advantages of multi dose study of drug for measurement of bioavailability over single dose study. 05
- D** Describe typical plasma drug concentration time profile in detail. 05
- E** Define bioequivalence. Discuss with advantages and disadvantages the Latin square cross-over design of bioequivalence in detail. 05
- F** Enumerate the factors influencing renal excretion of drugs and explain any two in brief. 05